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Invited Review Article

Beyond symptoms: Unlocking the potential of coronary calcium scoring in the prevention and treatment of coronary artery disease

Toufik Abdul-Rahman [a], Zarah Sophia Blake Bliss [b], Ileana Lizano-Jubert [b], Maria Jimena Salas Muñoz [b], Neil Garg [c], Vamsi Krishna Pachchipulusu [d], Patrick Ashinze [e], Goshen David Miteu [f], Rusab Baig [g], Dhuha Abdulraheem Omar [h], Marwa M. Badawy [i], Syed Muhammad Awais Bukhari [j], Andrew Awuah Wireko [a], Abdullahi Tunde Aborode [k], Oday Atallah [l], Hassan A. Mahmoud [m], Wesam Aldosoky [n], Shady Abohashem [n,*]

[a] Medical Institute, Sumy State University, Sumy, Ukraine
[b] Faculty of Medicine, Anahuac University North Campus, Huixquilucan, Mexico
[c] Rowan-Virtua School of Osteopathic Medicine, Stratford, New Jersey, United States
[d] Department of Internal Medicine, St Martinus University, Curacao
[e] Department of Medical Services, Saint Francis Catholic Hospital, Okpara Inland, Delta, Nigeria
[f] Department of Biomedical Science, School of Medicine and Health Sciences, University of North Dakota, Grand Forks, ND, United States
[g] Shadan Institute of Medical Sciences, Hyderabad, India
[h] Ministry of Health, Kurdistan Regional Government, Erbil, Iraq
[i] Faculty of Medicine October 6 University, Giza, Egypt
[j] Nishtar Medical University, Multan, Pakistan
[k] Department of Chemistry, Mississippi State University, Starkville, United States
[l] Department of Neurosurgery, Hannover Medical School, Carl-Neuberg-Strasse 1, 30625 Hannover, Germany
[m] Sherwood Forest Hospitals, NHS Foundation Trust, United Kingdom
[n] Massachusetts General Hospital and Harvard Medical School, Boston, United States

A R T I C L E I N F O

Central Illustration: Summary of CAC score test in CAD
This illustration highlights the indications, clinical benefits, and limitations of the CAC scoring test in CAD.

A B S T R A C T

Coronary Artery Disease (CAD) represents a persistent global health menace, particularly prevalent in Eastern European nations. Often asymptomatic until its advanced stages, CAD can precipitate life-threatening events like myocardial infarction or stroke. While conventional risk factors provide some insight into CAD risk, their predictive accuracy is suboptimal. Amidst this, Coronary Calcium Scoring (CCS), facilitated by non-invasive computed tomography (CT), emerges as a superior diagnostic modality. By quantifying calcium deposits in coronary arteries, CCS serves as a robust indicator of atherosclerotic burden, thus refining risk stratification and guiding therapeutic interventions. Despite certain limitations, CCS stands as an instrumental tool in CAD management and in thwarting adverse cardiovascular incidents. This review delves into the pivotal role of CCS in CAD diagnosis and treatment, elucidates the involvement of calcium in atherosclerotic plaque formation, and outlines the principles and indications of utilizing CCS for predicting major cardiovascular events.

* Corresponding author at: Cardiovascular Imaging Research Center - Radiology Department Massachusetts General Hospital and Harvard Medical School. Office: 165 Cambridge St, Suite 400, Boston, MA 02114-2750, United States.
E-mail address: SAbohashem@mgh.harvard.edu (S. Abohashem).

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Abbreviations

ACC	American College of Cardiology
AHA	American Heart Association
ASCVD	Atherosclerotic Cardiovascular Disease
CAD	Coronary Artery Disease
CCS	Coronary Calcium Scoring
CIMT	Carotid Intima-Media Thickness
CT	Computed Tomography
CVD	Cardiovascular Disease
CRP	C-Reactive Protein
JCS	Japanese Circulation Society
MACE	Major Adverse Cardiovascular Events

Introduction

Coronary Artery Disease (CAD) ranks as the third leading cause of death globally, with an estimated 126 million individuals afflicted worldwide. Eastern European nations bear a disproportionately high prevalence of this malady.[1,2] CAD manifests in a spectrum of conditions, from stable angina to myocardial infarctions, primarily driven by coronary artery occlusion stemming from atherosclerotic plaque accumulation, subsequently compromising myocardial oxygen supply.[3] Although a progressive disorder, entailing a complex interplay of risk factors and molecular pathways, CAD often remains asymptomatic until its advanced phases—especially in diabetic populations—culminating in potentially fatal events like myocardial infarction or stroke.[4] Well-documented is the prognostic utility of calcium deposition within coronary arteries, wherein elevated coronary calcium scores (CCS) correlate with a heightened CAD risk and unfavorable cardiovascular incidents.[5]

The CCS, or Agatston score, acquired via non-invasive computed tomography (CT), has gained traction as a cost-effective, reproducible, and steadfast assay for CAD risk appraisal.[5] Boasting superior validation in gauging coronary artery calcium (CAC) relative to other techniques, CCS remains the clinical gold standard for CAC quantification.[6] CCS computation entails the multiplication of calcium density (measured in Hounsfield units), facilitating the stratification of patients into low, moderate, or severe risk categories for major cardiovascular outcomes.[7] Regulatory guidelines across the United States and Europe endorse not only the CCS as a robust cardiovascular risk evaluation tool but also its potential in directing preventive therapy selection.[8,9] Emerging evidence

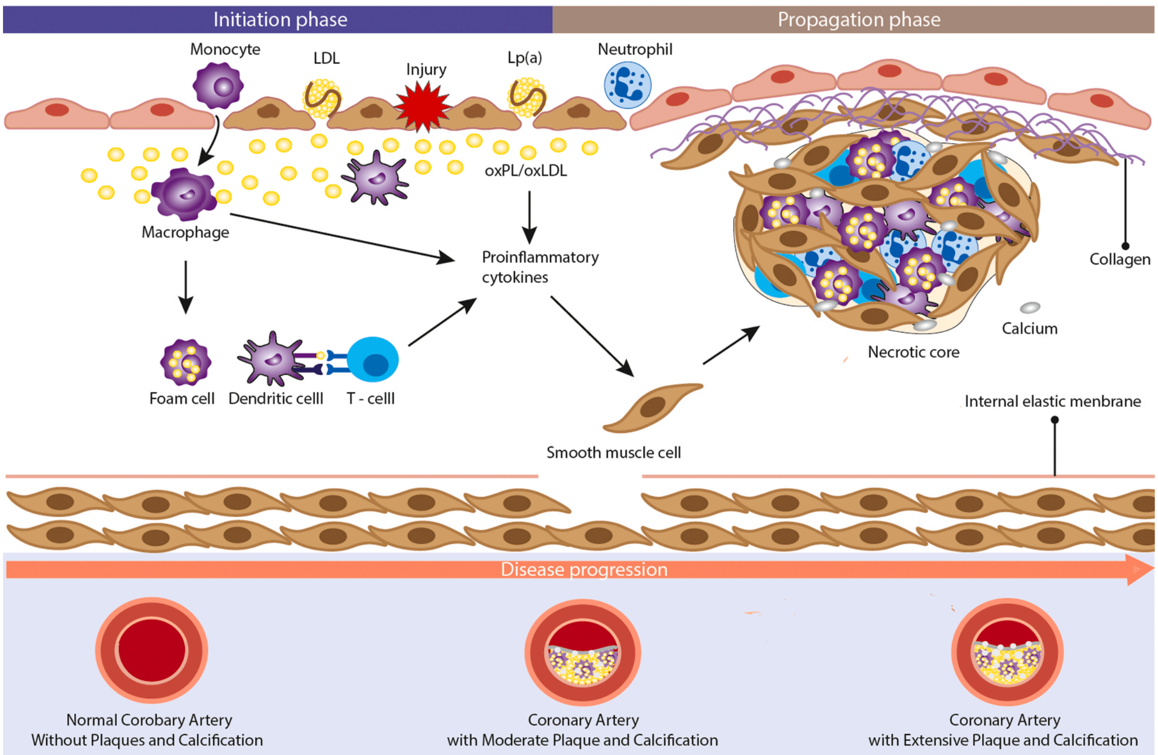

Fig. 1. Mechanism of plaque formation and calcification.

underscores that a CAC-score-centric approach could refine patient selection for statin therapy, thereby enhancing clinical outcomes vis-à-vis risk-factor-based strategies.[10] Furthermore, CAC scoring potentially informs non-pharmacological interventions targeting lipid and glucose management, enabling precise tailoring of therapeutic regimens to mitigate adverse cardiovascular event risks.[11] Despite the burgeoning literature on CCS's predictive prowess in cardiovascular risk, a paucity of investigations delves into its utility in orchestrating CAD treatment decisions. This review endeavors to bridge this knowledge gap by furnishing a comprehensive exploration of CCS's role in therapy selection and CAD management.

Role of calcium in plaque formation and CAD development

Calcium, an abundant and physiologically pivotal mineral in the human body, underpins the functional integrity of numerous cellular and organ systems.[12] While majorly deposited in teeth and bones, about 1% of calcium circulates in the bloodstream, playing a key role in vascular health.[13] With advancing biological age, dystrophic calcification—a pathophysiological alteration leading to calcium deposition in deteriorating tissues—can ensue, unfavorably impacting vascular architecture.[14] Evidence underscores calcification's cardinal role in atherosclerotic plaque formation, a seminal event precipitating adverse coronary incidents.[14] Atherosclerotic plaque formation, illustrated in Fig. 1, epitomizes a chronic degenerative cascade centering around calcium deposition in arterial walls, which consequently induces vessel narrowing, manifesting in pathologies such as stenosis, ischemia, and CAD.[14]

The intertwining of calcium with cardiovascular disease (CVD), the predominant mortality cause in the U.S., has been extensively investigated given that vascular calcification is discerned in a staggering 90% of CVD-afflicted patients.[15] Intriguingly, the pathogenesis of vascular calcification mirrors the dynamic machinery of bone remodeling, orchestrated by osteoblasts, osteoclasts, cytokines, hormones, and minerals.[16] Excessive bone turnover triggers bone degradation, thereby releasing surplus calcium, some of which may accrete in the vasculature, fostering atherosclerosis and plaque formation. This dysregulation, often age-associated, may manifest as osteoporosis in certain individuals.[15,16] Historically, CAD's genesis was predominantly attributed to passive lipid deposition within the vasculature, culminating in plaque formation.[14,17] However, recent advances reveal a nuanced yet profound interlink between CAD and calcification, fortifying the notion of a calcific atherogenesis.[14] Histological and molecular corroborations further cement this link, evincing a robust correlation between plaque burden and the extent of calcium deposition.[16]

While the majority of atherosclerotic plaques contain calcium, it's crucial to recognize that not all plaques are calcified, and a lack of calcium does not conclusively rule out the presence of CAD. This underscores the complexity of plaque composition and the need for comprehensive diagnostic approaches.

Caption: The process of plaque formation begins with the deposition of oxidized lipids into the intima of vessels following endothelial injury. This initiates the recruitment of monocytes to the intima. Monocytes differentiate into macrophages at the intima taking up oxidized lipids to form foam cells. In addition, macrophages secrete inflammatory cytokines triggering an inflammatory response and mineralization of extracellular matrix. The accumulation of foam cells together with the proliferation of smooth muscle cells leads to the growth of plaque while a stiffer matrix results in the formation of calcific minerals.

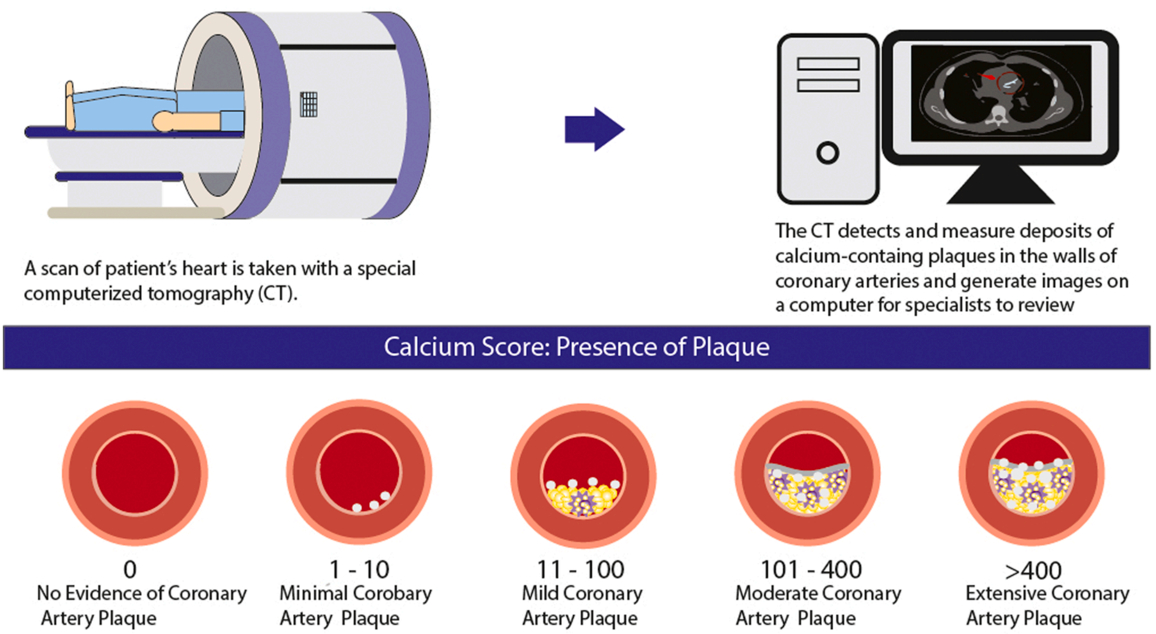

Fig. 2. The Concept of CCS Scan.

Principle of coronary calcium score

The quantification of CAC is endorsed for individuals perceived to be at risk of developing CAD, yet remain asymptomatic.[18] This technique is lauded for its reliability, reproducibility, and predictive accuracy in gauging cardiovascular risk.[18] The CCS procedure harnesses computed tomography, a non-invasive diagnostic modality, to scan the cardiac region, quantify calcium deposits within the coronary arterial walls, and generate interpretable images (Fig. 2).[19] Currently, the electrocardiogram-gated multidetector computed tomography scanning stands as the gold standard for this procedure.[19] The acquisition of CAC data, achievable within a few cardiac beats, is optimally conducted during the mid-systole phase of the cardiac cycle.[19]

Subsequent to image capture, a quantitative analysis is undertaken to score based on the density and dimensions of calcific lesions. Each calcific lesion's area is multiplied by a corresponding weighting factor (ranging from 1 to 4), with the total CAC score derived by summing the scores of individual calcified lesions.[19] These scores are further stratified to account for variations in sex and race.[19] The CAC scores are categorized as elucidated in Fig. 2:

0: Absence of plaque, signifying a very low likelihood of significant CAD.

1-10: Minimal plaque, indicating a low probability of significant CAD.

11-100: Mild atherosclerotic plaque, denoting mild to minimal likelihood of coronary artery stenosis.

101-400: Moderate atherosclerotic plaque, suggesting a high probability of non-obstructive CAD, with potential for obstructive disease.

400: Extensive atherosclerotic plaque, indicating a high likelihood of coronary artery stenosis

Caption: During a CCS test, a scan of a patient's heart is taken with a CT machine which measures the amount of calcium present in the walls of coronary arteries. The results of the scan are often shown on a computer as numbers called the Agatston score.

Current Clinical Guidelines for CAC

A significant number of CVD-related deaths occur without prior cardiac symptoms or diagnosis, emphasizing the need for a highly sensitive risk assessment tool, such as CAC.[20,21] This is particularly crucial considering that the most reliable marker for CV health is the overall burden of atherosclerosis.[21] Although guidelines for utilizing CAC in CAD risk assessment and treatment guidance vary globally, their ultimate aim remains consistent: reduce mortality and morbidity through apt risk stratification.[20] The broad consensus endorses CAC utilization to efficaciously stratify individuals with an intermediate 10-year atherosclerotic cardiovascular disease (ASCVD) risk, thus illuminating the necessity of commencing cardioprotective medications and lifestyle amendments.[22] Therefore CAC emerges as a personalized decision-making tool, empowering healthcare providers and patients to deliberate on the risks and merits entailed in initiating lifelong preventive therapy.[21]

Indications for CAC Scoring

The utilization of CAC scoring is governed by established guidelines that correlate the score with a patient's 10-year ASCVD risk, categorized as <5% for low risk, 5-20% for intermediate risk, and ≥20% for high risk.[23] The discourse surrounding high-risk cohorts in these guidelines is minimal as their symptomatic manifestation often directly leads to preventive therapy initiation, rendering CAC scoring unnecessary.[20] In contrast, the guidelines predominantly emphasize the intermediate-risk cohorts, where CAC scoring provides a robust mechanism for physicians to reclassify these patients effectively. The Society of Cardiovascular Computed Tomography-SCCT advocates for the use of CAC scoring in asymptomatic individuals aged 40-75 who are at intermediate risk, as well as those with a familial predisposition to premature CAD, irrespective of a <5% risk.[21] Leading global institutions such as the American College of Cardiology/American Heart Association (ACC/AHA), the European Society of Cardiology, the Endocrine Society, and the National Lipid Association have bestowed a class IIa recommendation upon CAC scoring, reinforcing its significance.[21]

A meticulous examination of risk-enhancing factors like dyslipidemia, familial history of premature ASCVD, inflammatory diseases, chronic kidney disease, and the like, is critical.[24] Should patients remain within the intermediate risk stratum post-assessment, the ambiguity surrounding the initiation of preventive therapy mandates the acquisition of a CAC score.[20,25,26] The guidelines from Australia particularly encourage the application of CAC for accurate risk stratification and subsequent follow-up in asymptomatic individuals aged 45-75 devoid of known CAD.[20] Similarly, the Canadian Cardiovascular Society reflects this stance but showcases a preference for statin treatment based on the Framingham Risk Score, allocating CAC scoring chiefly for asymptomatic individuals aged ≥40 within the intermediate risk bracket.[27] Over in the UK, the National Institute for Health and Care Excellence (NICE) integrates CAC scoring, with an added emphasis on evaluating the relevance of electrocardiographic alterations, especially ischemia in asymptomatic individuals.[28] Additionally, NICE routinely employs CAC scoring for meticulous risk stratification in diabetic patients, irrespective of whether they have type 1 or type 2 diabetes, in the absence of other risk factors.[28]

The global consensus for individuals with a <5% 10-year ASCVD risk (low-risk) is to consider CAC scoring when risk-enhancing factors are present along with a known family history of premature CVDs.[21] The CCS delineates a threshold of >40 years of age, with additional considerations given to genetic predispositions like familial hypercholesterolemia and elevated lipoproteins.[27] Australian guidelines resonate with this outlook but extend the indication to low-risk patients aged 40-60 with diagnosed diabetes mellitus.[29]

Asian regulatory bodies, notably the Chinese Society of Cardiology and the Japanese Circulation Society (JCS), exhibit caution in universally endorsing CAC. China principally employs CAC as a determinant for aspirin utilization.[20,30,31] Although JCS recognizes the prognostic value of CAC, especially in intermediate to high-risk cohorts, it calls for a stronger evidentiary basis from local studies to justify its clinical use, attributing this stance to the relatively lower CVD morbidity and mortality rates in Japan compared to other nations.[30,31-32] This cautious approach arises despite the burgeoning prevalence of CVD as a major mortality driver in Asia over recent years.[33]

A unanimous endorsement among global guidelines is the suitability of CAC testing for individuals aged 40–75 with a 5–20 % 10-year ASCVD risk. Moreover, CAC testing is deemed suitable in scenarios where a family history of premature CAD is discernible even if the risk is categorized as low.[22] This shared global stance exemplifies the critical role of CAC scoring in augmenting the clinical decision-making process, especially among the intermediate-risk cohorts, thereby fortifying the collaborative strategy to mitigate cardiovascular risks.

Treatment Protocols Based on CAC Score

The CAC score plays a crucial role in determining the appropriate treatment option. Its purpose is to facilitate a collaborative decision-making process between the patient and physician regarding initiating preventive therapy. Various CAC ranges have been established to guide treatment decisions. In line with global guidelines, a CAC score of zero generally does not qualify for statin therapy unless the patient has not achieved their LDL-C goals as per European guidelines.[20] Conversely, American, and Canadian guidelines indicate that a CAC score ranging from 1 to 99 may not necessarily require statin implementation. However, it may be a reasonable choice for individuals over 55 years, based on the discretion of the patient and physician. Meanwhile, Oceanian guidelines favor the adoption of lifestyle modifications within this range. If a patient's CAC score exceeds 100, European, Canadian, and American guidelines unanimously recommend the initiation of pharmacotherapy. In contrast, Australian and New Zealand guidelines suggest considering statin therapy for CAC scores ranging from 101 to 400 only if the score exceeds the 75th percentile for age or if the CAC score surpasses 400.[21,26,29]

When should CAC be repeated?

The timing for repeating the CAC test varies globally, with specific guidelines depending on the initial CAC score. For individuals with a CAC score of 0, different rules apply based on risk level. In low-risk individuals, it is generally recommended to repeat the test in 5-10 years. For intermediate-risk patients, a repeat test is advised after five years, while high-risk patients or those with diabetes are recommended to have a repeat test in 3 years.[21] Oceanian guidelines suggest repeating CAC scores every three years if the values fall between 101 and 400. However, if the values exceed 400, repeating the CAC test is unnecessary as it would not affect the physician's clinical decision.[29] The National Lipid Association recommends a rescan every three years for patients with a CAC score greater than 100 and diagnosed with dyslipidemia.[34] For patients with a previous CAC score ranging from 1 to 100, the decision to rescan should be based on clinical judgment.[21] It is crucial, however, to consider the cumulative exposure to ionizing radiation from repeated CT scans, particularly for individuals requiring frequent monitoring.

Clinical significance

The clinical significance of utilizing CAC in patients with CAD is substantial, offering several notable benefits. Early management of CVD has demonstrated enhanced cost-effectiveness and improved outcomes in terms of morbidity and mortality. Given the direct correlation between CAC scores and the severity of CAD and future coronary events, CAC indicates an association with ASCVD rates of ≥7.5/1000 person-years.[35] Detecting significant CAC values at an early stage enables proper risk stratification and facilitates initiating appropriate management strategies.[36] Moreover, research has revealed that incorporating CAC scores results in significant improvements in risk reclassification.[35] Additionally, patients who have a clear understanding of their CAC score are more likely to adhere to their prescribed medication regimen and make necessary lifestyle modifications to support their overall health.[21]

CAC as a predictor of CAD risk

Research insights into CAC as a CAD risk predictor

The advent of CAC scanning for CVD risk prediction heralded a notable evolution in shared clinical decision-making processes. Since its initial description by Agatston et al. in 1990 as a non-contrast-enhanced, electrocardiographically gated CT method to quantify CAC, multiple studies have fortified the robust association between CAC and cardiovascular outcomes in asymptomatic individuals.[37,38] The validation of CAC as a pivotal tool for stratifying CVD risk among middle-aged and older adults is extensively documented. However, the 2019 guidelines by the American College of Cardiology/American Heart Association posited a requisite for additional data to thoroughly evaluate the efficacy of CAC in the risk assessment of younger adults deemed low risk.[39]

A comprehensive cohort study spanning 1997 to 2009, encompassing 13,397 patients aged 30-49 years devoid of CVD or malignancy, divulged the augmentation of prognostic accuracy for major adverse cardiovascular events (MACE) and all-cause mortality by CAC.[39,40] The presence of any CAC was coupled with elevated long-term hazards of MACE and MI, whereas severe CAC amplified hazards across all outcomes, including death. This suggests a potential utility of CAC in guiding clinical decision-making among select young adults.[39]

The most expansive cohort examining CAC, known as the CAC Consortium, was conducted in 2016 to probe the association between CAC and long-term cause-specific mortality, including CVD and non-CVD mortality.[37] With 66,636 asymptomatic adult participants devoid of known CVD at baseline, recruited between 1991 and 2010 from four US study sites, CAC testing formed an integral part of their clinical CVD risk assessment. The subsequent categorization of patients into high, intermediate, and low-risk groups for all-cause mortality based on CAC scores illuminated its seminal role in risk stratification.[37]

In a 2023 study, Alexander et al. examined 2,869 primary prevention patients with a CAC score greater than 1000, exploring the relationship between risk markers and ASCVD mortality over a median period of 11.8 years. Their findings underscored diabetes and severe left CAC as independent prognostic indicators of heightened risk for this patient group.[41] Interestingly, although a zero CAC score had a 99.9% negative predictive value for obstructive coronary artery stenosis and combined outcomes like death, myocardial infarction, or coronary revascularization, Michael et al. (2023) noted that 7% of these patients presented with noncalcified plaque. This suggests the need for prudence before reducing preventive measures in patients with chest pain, a zero CAC score, and multiple risk factors.[42] Further, recent literature points to the efficacy of CAC scoring in evaluating cardiovascular risk among specific cohorts, including those with inflammatory bowel disease, type 2 diabetes, and atrial fibrillation patients undergoing ablation.[43–45]

Beyond the traditional risk calculators and non-invasive tests, CAC has displayed a consistent prowess in cardiovascular risk evaluation, particularly among patients categorized as intermediate risk. As delineated by Dustin M. Thomas et al., patients with CAC $= 0$ exhibit a markedly low risk of future cardiac events, while those with CAC ≥ 100 face a substantial coronary disease risk.[46]

Comparative verdict: CAC vis-à-vis other CAD risk predictors

Risk Prediction Models (RPMs), epitomized by their capacity to estimate CAD risk based on an array of predictive clinical variables, stand as invaluable instruments to steer cardiovascular risk management. By identifying individuals likely to garner the most benefit from various interventions, RPMs play a pivotal role in personalized medicine.[47] These encompass Carotid Intima-Media Thickness (CIMT), brachial flow-mediated dilation, ankle-brachial index, high-sensitivity C-Reactive Protein (CRP), family history of disease, and CAC.[48]

A seminal comparison by A Yeboah et al. (2012) posited CAC as the zenith of risk markers in augmenting discrimination, prediction, and reclassification of cardiovascular risk, particularly among individuals classified as intermediate risk.[48] Subsequent studies have independently endorsed CAC as a robust risk estimation instrument, outperforming the Framingham estimator, biomarker approach, and other risk estimators in predictive capacity.[21,49] Although clinically pertinent, biomarkers like CRP and CIMT fall short of the predictive efficacy of CAC scores, especially when utilized alongside risk scores.[21] The Framingham coronary artery heart risk estimator, despite its widespread adoption and proven viability, exhibits clinical limitations in delineating the precise subset of individuals likely to encounter cardiovascular complications.[50,51,52] However, studies have found that prognosis models based on solely risk factors have clinical limitations in the delineation of who will or will not experience cardiovascular complications.[53]

The incorporation of CAC scoring using computed tomography is a commendable stride toward ameliorating predictive accuracy. A plethora of studies underpin the hypothesis that elevated CAC scores can markedly modify and enhance risk prediction gleaned from the FRS alone, particularly among intermediate-risk individuals.[53] A meticulous meta-analysis in a NEJM general medicine study scrutinized the incremental merits of including CAC scoring in cardiovascular risk prediction vis-a-vis traditional risk scoring alone.[54] The meta-analysis, encapsulating six studies with varied designs, ascertained that CAC scoring modestly augments traditional risk scores for CVD prediction.[54] The reclassification of individuals from low to intermediate/high risk, and vice versa, post-CAC scoring, underscored a significant alteration in risk stratification and demonstrated an imperative role of CAC in clinical decision-making.[54]

In summation, Coronary Artery Calcification imaging is heralded as the quintessential cardiovascular risk marker for asymptomatic individuals. Its potential to steer clinical decision-making and augment the robustness of patient profiles beyond traditional risk factors is unparalleled. The cost-effectiveness and predictive superiority of CAC, especially in preventive strategies, markedly outweigh surrogate approaches, asserting its pivotal role in the landscape of CVD management.[54]

CAC-guided treatment strategies for CAD

CAC scoring is not only instrumental in risk stratification among individuals with suspected or established CAD but is also pivotal in steering treatment decisions. It facilitates the identification of individuals poised to benefit from preventive or aggressive medical therapies.

The merit of CAC assessment as a tool to refine CAD treatment and ameliorate cardiovascular health outcomes is particularly pronounced in individuals with a familial history of premature coronary disease. The CAUGHT-CAD study underscored that a statin treatment protocol steered by CAC scores ranging from 1 to 400 could galvanize both patients and healthcare providers to enhance risk factor control and attenuate cardiovascular risk.[55] While CAC scoring can evaluate CAD risk in both asymptomatic and symptomatic individuals, it's noteworthy that most guidelines deter the use of CAC scoring to exclude obstructive CAD in symptomatic patients. Its utility in shared decision-making concerning statin use for individuals at intermediate risk of CAD is recognized, albeit not endorsed as a solitary screening tool or for management predicated solely on CAC score.[56]

Venkataraman et al. (2021) explored the ramifications of a statin treatment protocol dictated by CAC assessment on cardiovascular risk and lifestyle behaviors over a 12-month period. Their findings suggest a positive influence on risk factor control among patients and clinicians, corroborating the potential benefits of CAC in primary prevention of CAD. Moreover, a substantial reduction in total cholesterol alongside moderate improvements in blood pressure was observed.[55] These patients, earmarked for statin treatment based on CAC scores between 1 and 400, were randomly allocated to either the intervention (CAC-guided) or control group in a 1:1 ratio.[55] Given that hypercholesterolemia is a notable independent predictor of cardiovascular mortality, lipid-lowering medications are associated with a favorable prognosis.[57] Although statin therapy is a cornerstone of lipid-lowering management and cardiovascular prevention, it doesn't benefit all patients. For such individuals, an array of non-statin therapies like bile acid sequestrants, fibrates, nicotinic acid, ezetimibe, bempedoic acid, volanesoren, evinacumab, and PCSK9 inhibitors are advocated as they demonstrate promise in lipid management and cardiovascular risk mitigation.[58]

Ajufo et al. (2021) dissected the association between CAC, bleeding, and ASCVD to evaluate the net estimated effect of aspirin at diverse CAC thresholds. Their findings unveiled a stronger association between higher CAC and ASCVD compared to bleeding events. The efficacy of aspirin in mitigating ASCVD events outweighed its propensity to induce bleeding only in groups with a high estimated 10-year ASCVD risk (>20%).[59] This aligns with the 2019 ACC/AHA guidelines on primary prevention of CVD, which dissuade the use of primary prevention aspirin in individuals at an elevated risk of bleeding, hinting at the potential of CAC scoring in pinpointing optimal candidates for low-dose aspirin in primary prevention.[59,60]

A Meta-Analysis by Gupta et al. (2017) found a positive association between CAC identification and the initiation or persistence of pharmacological and lifestyle preventive therapies. Individuals with discernible CAC were 2-3 times more likely to embark on preventive pharmacological therapy (encompassing aspirin, lipid-lowering, and blood pressure-lowering drugs) and adopt exercise or dietary modifications compared to those devoid of CAC.[61]

Lastly, Parcha et al. (2017) conjectured that CAC scoring could steer the initiation of hypertension therapy and personalized preventive strategies to diminish cardiovascular risk, especially in cohorts where current guidelines are non-committal on treatment recommendations. This personalized risk reduction, particularly among individuals with CAC > 0, may encompass antihypertensive therapy, lipid-lowering therapy, glycemic control, a nutritious diet, physical activity, and smoking cessation. Integrating CAC score–guided cardiovascular risk reduction with intensive BP control, especially in individuals presumed to be at low cardiovascular risk with no pharmacotherapy recommendation under the prevailing guidelines, could unfold as a cost-effective strategy.[62]

Challenges and limitations of calcium scoring

Despite its utility, CAC scoring has several limitations and potential disadvantages. One notable concern is the exposure to ionizing radiation during the CT scans used for CAC scoring. While the radiation dose is typically low, repeated scans over time may increase cancer risk.[63] Another limitation is the possibility of false positives and false negatives in CAC scores, which can lead to unnecessary interventions or a false sense of security.[64] These inaccuracies may include non-calcified plaques, calcifications in bypass grafts, limited image resolution, and interpretive errors. The cost of CAC scoring may also be a barrier to its use, as it may not be covered by insurance and can be expensive for patients paying out-of-pocket. Moreover, the applicability of CAC scoring may be limited in certain populations, such as younger individuals with low CAD risk or older individuals with high CAD risk. Another significant limitation is the discrepancies in risk scores reported between low and high-risk populations using the Framingham risk equation.[53,65] Risk scores may be overestimated in low-risk populations and underestimated in high-risk populations, potentially resulting in inappropriate therapy selection. Additionally, the current standard for CAC risk scores may not apply to all individuals due to differences in body composition and variations among individuals. While the scores may predict the general population's risk, they do not capture personalized discernment of an individual's risk.[53]

Another limitation of CAC scoring is the lack of consensus regarding the optimal score for predicting CAD in different populations. Elderly individuals, young adults, athletes, and people with diverse lifestyles may have varying scores, making it challenging to establish a universally applicable threshold for CAD risk prediction. Furthermore, studies examining the relationship between physical activity and CAC have yielded inconsistent or inconclusive results.[66] Conflicting findings arise due to differences in age, sex, cardiovascular risk factors, and methodologies employed to measure physical activity. This underscores the need for a personalized approach to risk assessment and therapy selection considering individual variations and lifestyle factors. Additionally, while CAC scoring improves risk stratification, there is limited evidence demonstrating its direct impact on clinical outcomes, such as reducing the incidence of myocardial infarction or stroke.[67] Consequently, it is important to exercise caution when interpreting CAC scores and to consider them in conjunction with other clinical and laboratory data, while also accounting for individual differences and lifestyle factors.

Future directions for CAC in CAD treatment guiding

While various screening methods have been evaluated, CAC testing has emerged as a leading contender due to its ability to provide a more accurate assessment of CAD risk.[68] When utilized correctly, CAC assessment can positively influence physician and patient risk factor control.[55] The presence of any CAC value significantly increases a patient's risk of CAD, highlighting the importance of proper management for these individuals to minimize future complications. Adding CAC to traditional risk factor assessments has been shown to improve the accuracy of patient management, particularly in guiding statin, aspirin, and antihypertensive therapies.

Current guidelines based on pooled cohort equations (PCE) endorse statin therapy in non-diabetic adults aged 40-70 with dyslipidemia and intermediate to high ASCVD risk.[20,24] However studies have demonstrated that PCE often overestimates the number of candidates for statin therapy, with approximately 41% of individuals having a CAC score of 0.[22,69] Incorporating CAC scoring in these cases can reclassify patients, reducing the likelihood of unnecessary adverse effects and improving the cost-effectiveness of therapy.[22,70] Most individuals in the overestimation group have intermediate ASCVD risk, as evidenced by the MESA study, which showed that CAC results did not alter statin therapy recommendations for low-risk and high-risk groups.[19,60] Additionally, symptomatic patients with a CAC score of 0 may still have obstructive CAD, making CAC alone unreliable in excluding CAD in this population.[19,71] However, CAC has demonstrated its ability to identify individuals at increased CVD risk and determine the potential benefit or harm of aspirin therapy.[60] In the context of antihypertensive therapy, current predictive equations provide only modest prognostic insight for risk stratification in low-risk stage I hypertension and do not assist in offering personalized management goals.[20,62,72] The role of CAC in hypertension treatment and management is still being explored. Still, it is hypothesized that by using the CAC score, physicians will be better equipped to identify patients requiring stricter blood pressure control and closer monitoring.[62]

Indeed, CAC serves as a valuable stratification tool that allows for better allocation of therapy and has the potential to identify individuals at high risk who may benefit from more aggressive treatment strategies. While tools like PCE can provide an initial risk ranking, CAC provides a visual assessment of the endothelium and the burden of atherosclerosis, a strong predictor of future cardiovascular events. By including CAC in patient analysis, it is possible to identify individuals who may be at higher risk than initially estimated by PCE alone. This improved risk ranking facilitated by CAC can guide therapy intensity, such as prescribing higher statins or including low-dose aspirin in the treatment regimen, thereby achieving optimal risk reduction. Moreover, CAC enables the appropriate stratification of patients with inherently high risk due to family history or previous medical diagnoses, including chronic kidney disease, diabetes mellitus, and familial hypercholesterolemia.[20]

Considering the economic burden of treatment is a crucial aspect when making healthcare decisions. A study conducted by Jonathan C. Hong et al. compared the cost-effectiveness of the CAC strategy with the PCE global guidelines strategy. The study found that in 51% of all simulations, the CAC strategy was the most cost-effective approach.[70] However, it is important to note that both strategies generally had similar clinical and economic consequences. While the study had some limitations, including the absence of a clinical trial, the overall conclusion was that CAC scoring was slightly more cost-effective and facilitated appropriate resource allocation and shared decision-making.[70] This suggests that integrating CAC scoring into treatment decisions can help optimize healthcare resource utilization and aid in discussions between healthcare providers and patients regarding the most suitable treatment options.

In summary, future directions for CAC in CAD treatment are refining risk assessment, optimizing patient management strategies, and investigating its potential in hypertension treatment. Incorporating CAC scoring into risk factor assessments can improve accuracy, personalize therapy, and mitigate unnecessary interventions, enhancing overall patient care.

Conclusions

In culmination, CCS, facilitated by computed tomography, has emerged as a reputable method for cardiovascular risk assessment, thereby playing an instrumental role in tailoring treatment strategies. Despite its invaluable potential in atherosclerosis management, CCS harbors certain limitations. Notable among these are variations in risk scores across low and high-risk cohorts, ambiguities concerning optimal scores within diverse demographic groups, and inconsistent correlation with physical activity. Nonetheless, for individuals predisposed to CAD, undergoing CCS is paramount as it furnishes substantial guidance for risk management and contributes significantly to the mitigation of adverse cardiovascular occurrences.

CRediT authorship contribution statement

Toufik Abdul-Rahman: Data curation, Visualization, Writing – review & editing. **Zarah Sophia Blake Bliss:** Writing – original draft. **Ileana Lizano-Jubert:** Writing – original draft. **Maria Jimena Salas Muñoz:** Writing – original draft. **Neil Garg:** Writing – original draft. **Vamsi Krishna Pachchipulusu:** Writing – original draft. **Patrick Ashinze:** Writing – original draft. **Goshen David Miteu:** Writing – original draft. **Rusab Baig:** Writing – original draft. **Dhuha Abdulraheem Omar:** Writing – original draft. **Marwa M. Badawy:** Writing – original draft. **Syed Muhammad Awais Bukhari:** Writing – original draft. **Andrew Awuah Wireko:** Writing – review & editing. **Abdullahi Tunde Aborode:** Writing – review & editing. **Oday Atallah:** Writing – review & editing. **Hassan A. Mahmoud:** Writing – review & editing. **Wesam Aldosoky:** Writing – review & editing. **Shady Abohashem:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

Authors wish to declare no conflict of interest.

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